

MultiExplorer - User Manual

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1 Installing

To use **MultiExplorer**, you will need the source code, **Docker**, and **XLauncher** (Windows version). For the CPU flow, a pre-compiled version of the **Sniper** simulator is required. **Java** will be required in the virtual machines flow.

1. **MultiExplorer**'s source code:

<https://github.com/lscad-facom-ufms/multiexplorer>

2. **Sniper**'s pre-compiled version:

<https://drive.google.com/file/d/1aXNxy60Z7NjP1XUgnh0GuTFAUePtwZkW/view>

3. **XLauncher**:

<https://sourceforge.net/projects/vcxsrv/>

4. **Docker**:

<https://docs.docker.com/desktop/install/windows-install/>

5. **Java**

https://download.oracle.com/java/22/latest/jdk-22_linux-x64_bin.deb

1.1 Source Code

MultiExplorer is open-source, and the code is freely available at <https://github.com/lscad-facom-ufms/multiexplorer>. Becoming a contributor is also possible.

1.2 Sniper

After downloading **Sniper**'s pre-compiled version, unzip the whole folder under the source code's root.

1.3 XLaucher

This tool will allow Docker Desktop for Windows to display MultiExplorer's graphical user interface. Download, install and launch it - you will have to go through a prompt wizard.

In step 1, select "Multiple windows" and set the "Display number" to -1 (unlimited).

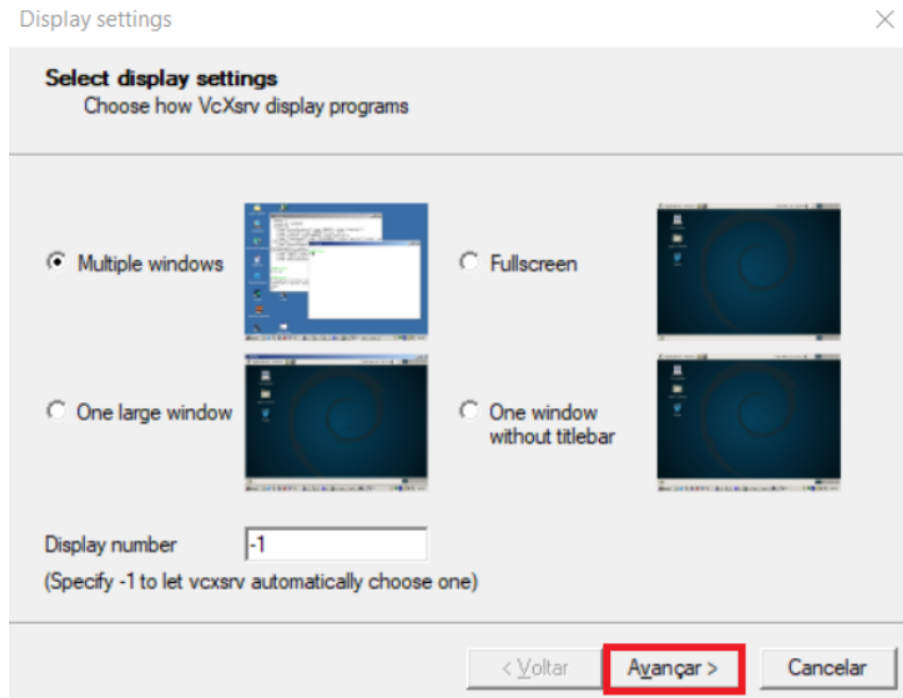


Figure 1: Select display settings. "Multiple windows" is recommended. Keep "Display number" value's at -1 to not limit the number of windows that can be launched.

In step 2, pick "Start no client" and click "Advance".

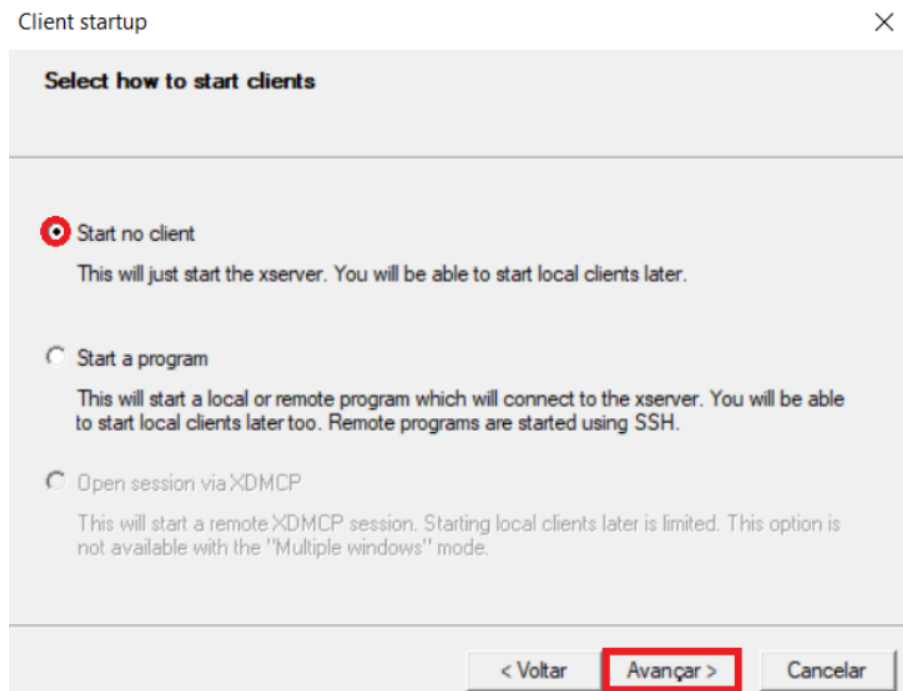


Figure 2: Select how to start clients. "Start no client" is recommended.

In step 3, check "Disable Access control". If you don't, the GUI won't work.

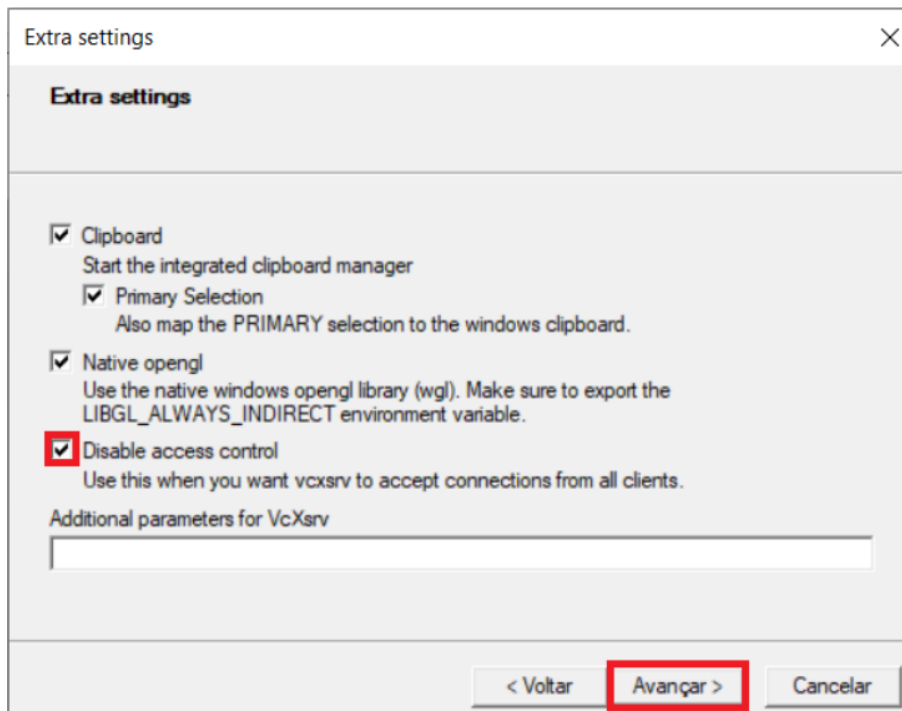


Figure 3: Extra settings. Disabling access control is required to use MultiExplorer's GUI from Docker.

In step 4, you can save your settings (optional) and finish this process.

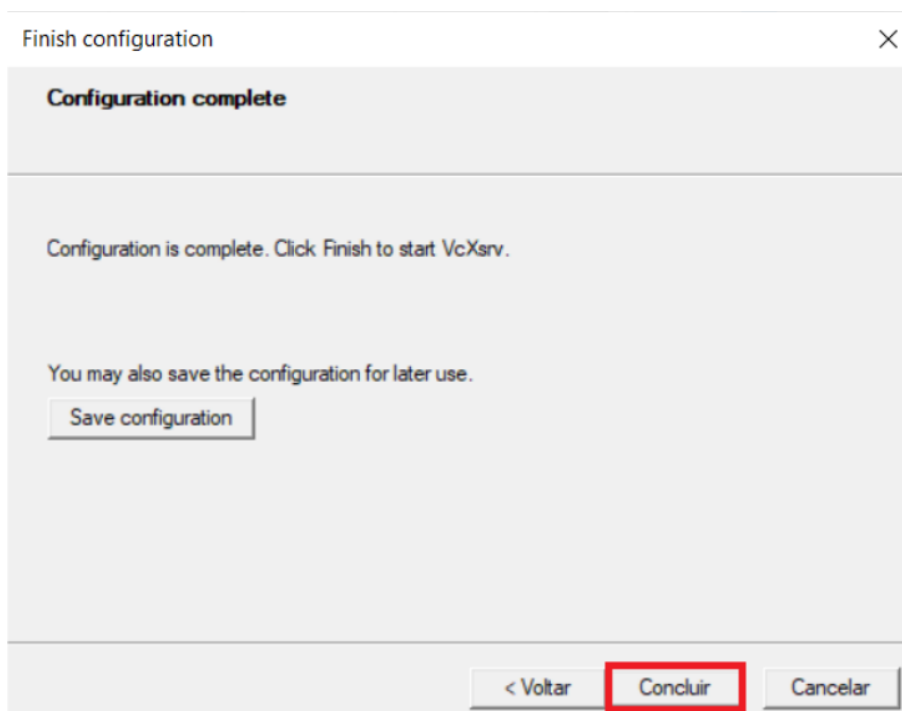


Figure 4:

XLancher must be running whenever you want to use MultiExplorer's GUI.

1.4 Docker

This tool allows the establishment, upkeep, and sharing of a homogeneous environment for the development, testing, and/or execution of an application. If you don't use it, you must manually set MultiExplorer's requirements in a Linux-based OS.

Once Docker is running, acquire your machine's local network address (IPV4):

```
ipconfig (1)
```

If there's no ".env" file in the source code's root folder, create it. This environment file must contain the following line:

```
DISPLAY="IPV4":0.0 (2)
```

Replace "IPV4" with your machine's local network address, acquired in the previous step. Afterwards, go to the source code folder:

```
cd path_to_multiexplorer (3)
```

Then, proceed with the following line command:

```
docker compose up -d (4)
```

This will build the preset Docker environment. Take note of the container's name. The next step is to run:

```
docker exec -it "container_name" bash (5)
```

Replace "container-name" with the actual container name generated by docker when the preset environment was built. The above line will start a command-line interface inside the preset Docker environment. Then, proceed with:

```
make install (6)
```

This line will use MultiExplorer's construction script (GNU Make ¹) to prepare the software requirements. The following line will finish the settings:

```
make config (7)
```

After this, the GUI should be ready to use. To launch it, XLaucher must be running, then you use the command:

```
python ME.py (8)
```

This way, the GUI should open, and you can start to use MultiExplorer.

1.5 Java

After downloading the **Java** JDK Debian package, placing it in the source folder, and installing it:

```
dpkg -i "debian-package-name".deb (9)
```

"debian-package-name" should be replaced with the debian package name.

¹<https://www.gnu.org/software/make/>

2 Using MultiExplorer

In this section we will approach how to make practical use of MultiExplorer.

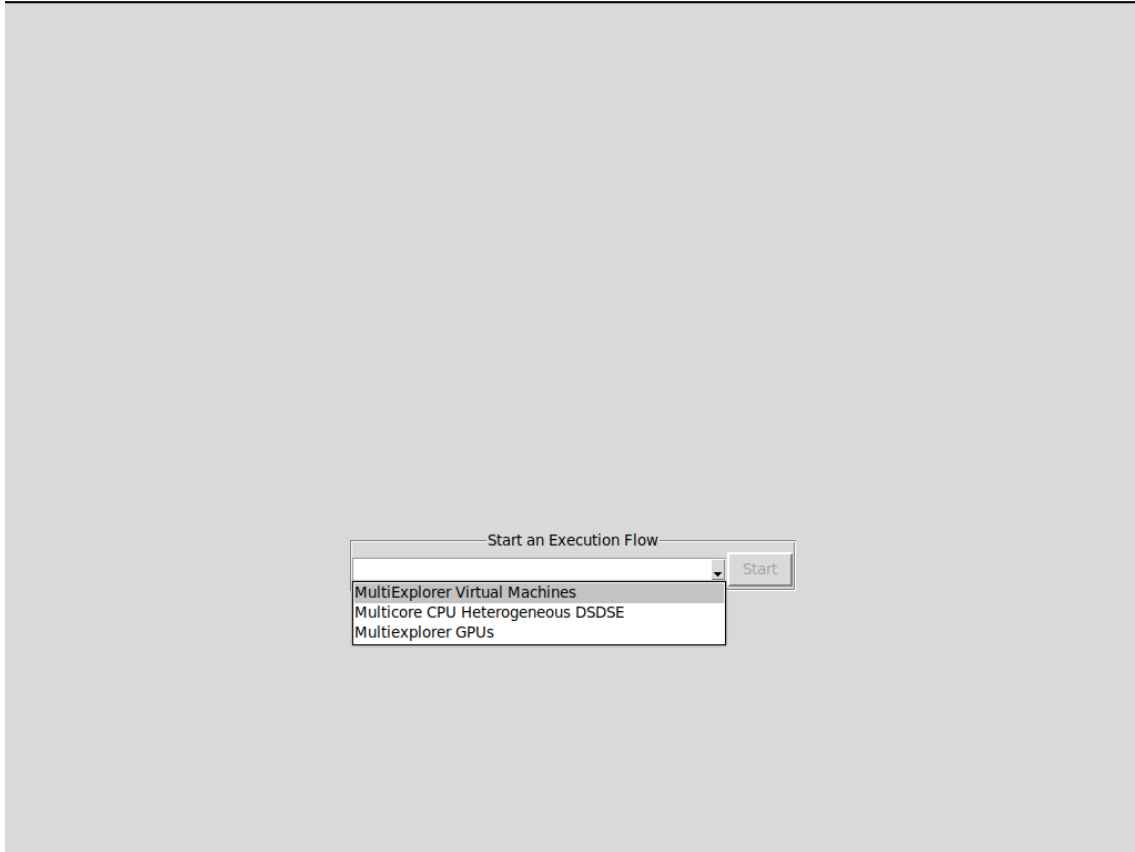


Figure 5: MultiExplorer’s initial window. Select an Execution Flow and press ”Start”.

In Figure 5, you will see that the first step is to select an ”Execution Flow”. There will be three options: ”MultiExplorer Virtual Machines”, ”Multicore CPU Heterogeneous DSDSE” and ”MultiExplorer GPUs”.

2.1 Multicore CPU Heterogeneous DSDSE

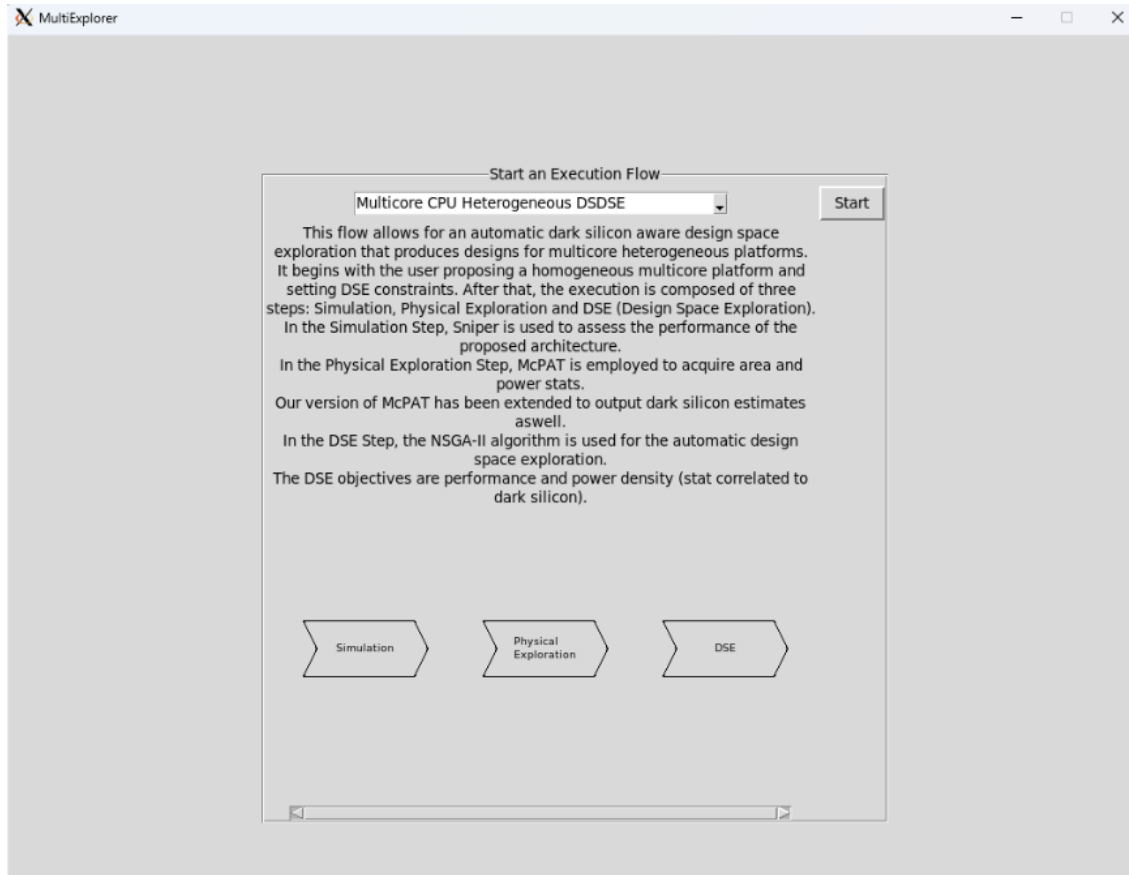


Figure 6: MultiExplorer’s CPU flow.

For the CPU flow, a brief description of this flow is first presented (Figure 6).

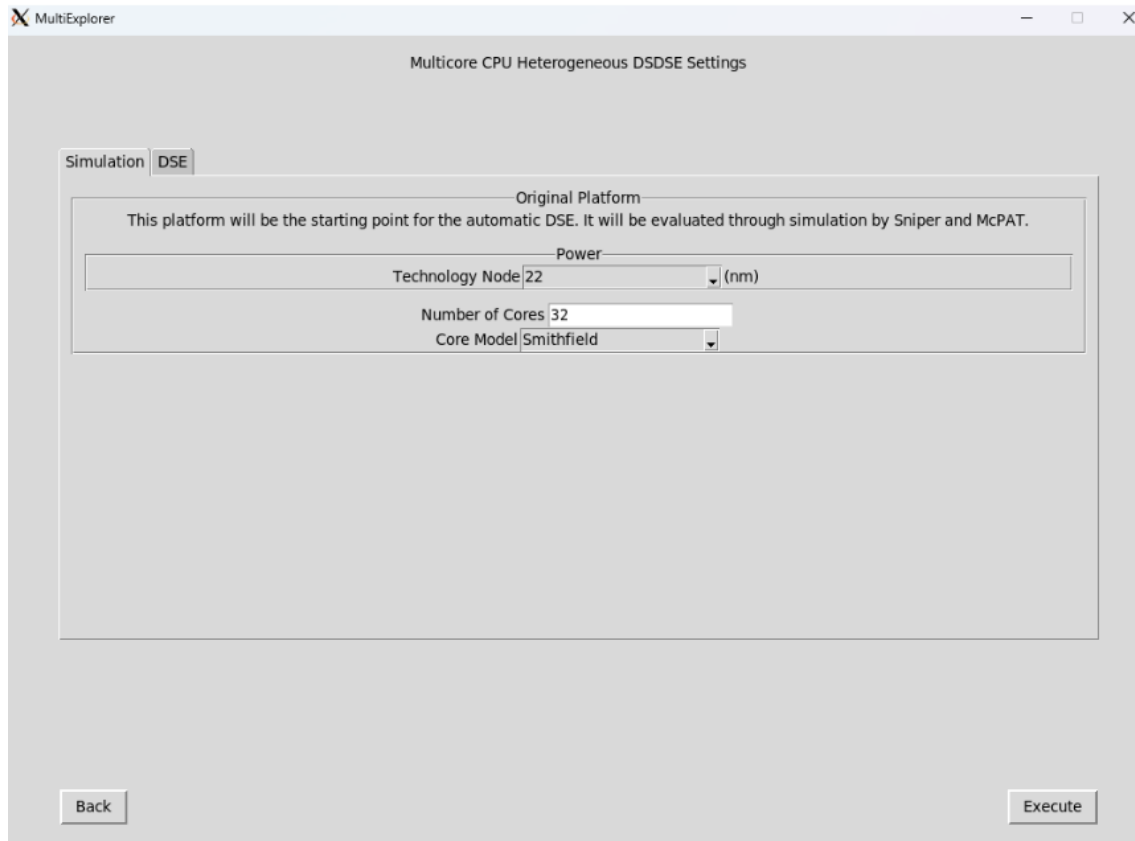


Figure 7: Platform description consist of technology node (lithograph), the number of cores, and the selection of a core model.

In figure 7, we can see the first user inputs. These inputs refer to the description of a homogeneous multicore platform. The user describes the platform in a high level of abstraction, merely selecting lithography, number of cores and core model. Each core model features a preset architecture specification that can be found in the folder "input-examples", in the form of JSON files.

MultiExplorer

Multicore CPU Heterogeneous DSDSE Settings

Simulation | DSE

Exploration Space

Define the limitations for core heterogeneity. The platforms produced by the automatic DSE will often contain two different core models.
You are able to restrict the number of cores used of both models.

Minimal number of cores from the original model - from 1 to 31

Number of cores that may be from another model - from 1 to 31

NSGA-II Parameters

Crossing Rate 50.0 (%)

Mutation Rate 10.0 (%)

Number of Generations 150

Population Size 10

Constraints

Maximum Power Density 175.25 (V/mm²)

Maximum Area 203.66 (mm²)

Back Execute

Figure 8: Design Space Exploration settings. Quantity of cores, NSGA-II calibration parameters, and DSE constraints are found here.

Figure 8 show a second tab of user inputs. In this tabe the user can calibrate the DSE algorithm, set goals and constraints.

The two inputs at the top are intervals (min-max) indicating how many cores the algorithm can allocate in a single multicore platform. Since the algorithm generates heterogeneous multicore platforms, you must set quantities for both the original core ("Core model", selected in the other tabe of user inputs), and for the alternative core model.

This tab also allows for some calibration parameters for NSGA-II to be changed. Usually, keeping the original parameters is recommended, unless you know what you are doing. Higher values for "Number of Generations" and "Population Size" can increase execution time exponentially.

Constraints for the generated multicore platforms are found at the bottom.



Figure 9: MultiExplorer’s Flow Execution. The progress of the simulations and design space exploration can be tracked in this window. Partial results might be available for each step finished.

After the user finishes the settings and press the "Execute" button, the tool will go to the execution window (figure 9). Here the progress can be tracked, and after all steps are done, the user can use the button "See Results" to navigate to the results’ window.

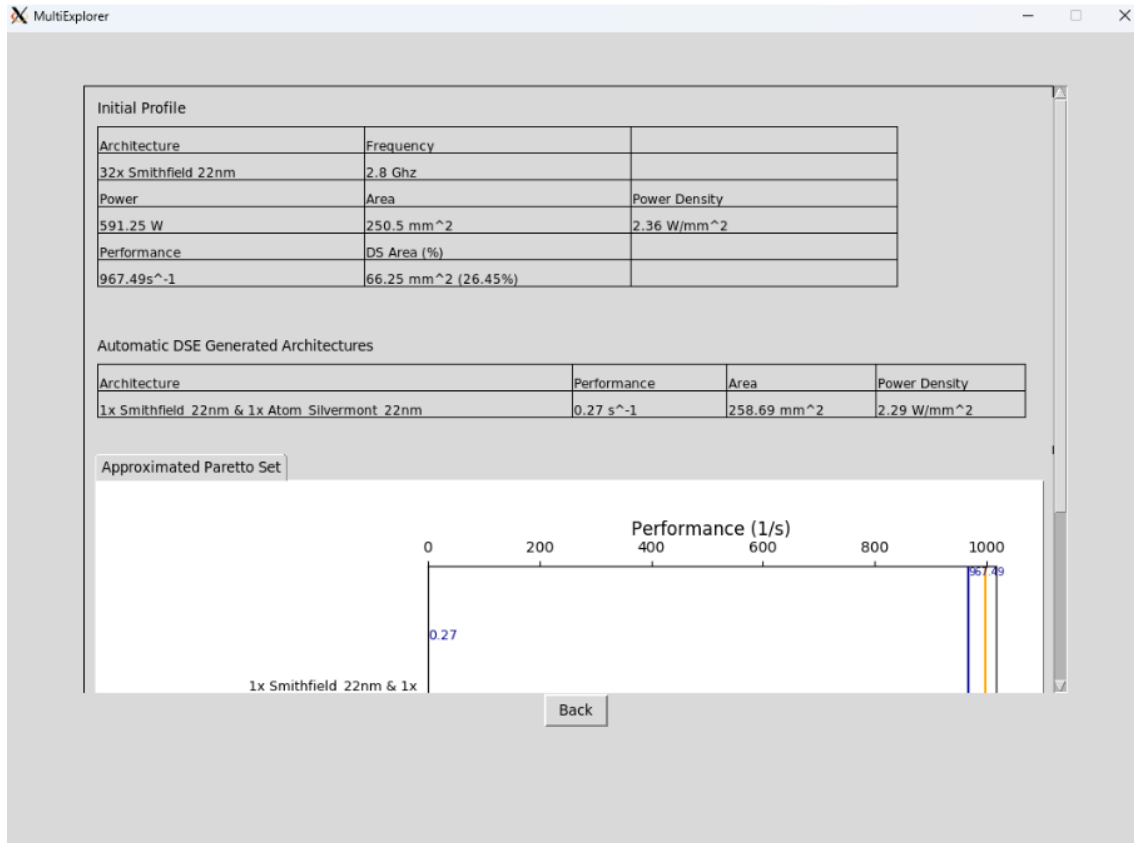


Figure 10: Results' Window. A profile of the user-specified homogeneous multicore platform can be found in the table at the top. Data on the heterogeneous multicore platforms generated by the algorithm can be found at the second table, and in the plot at the bottom.

The plot in the result's window (figure 10), presents performance and power density for each heterogeneous multicore platform in the approximated Pareto set found by the algorithm. These settings and results are also available in the form of files in the folders under "/rundir/Multicore CPU Heterogeneous DSDSE".

2.2 MultiExplorer Virtual Machines

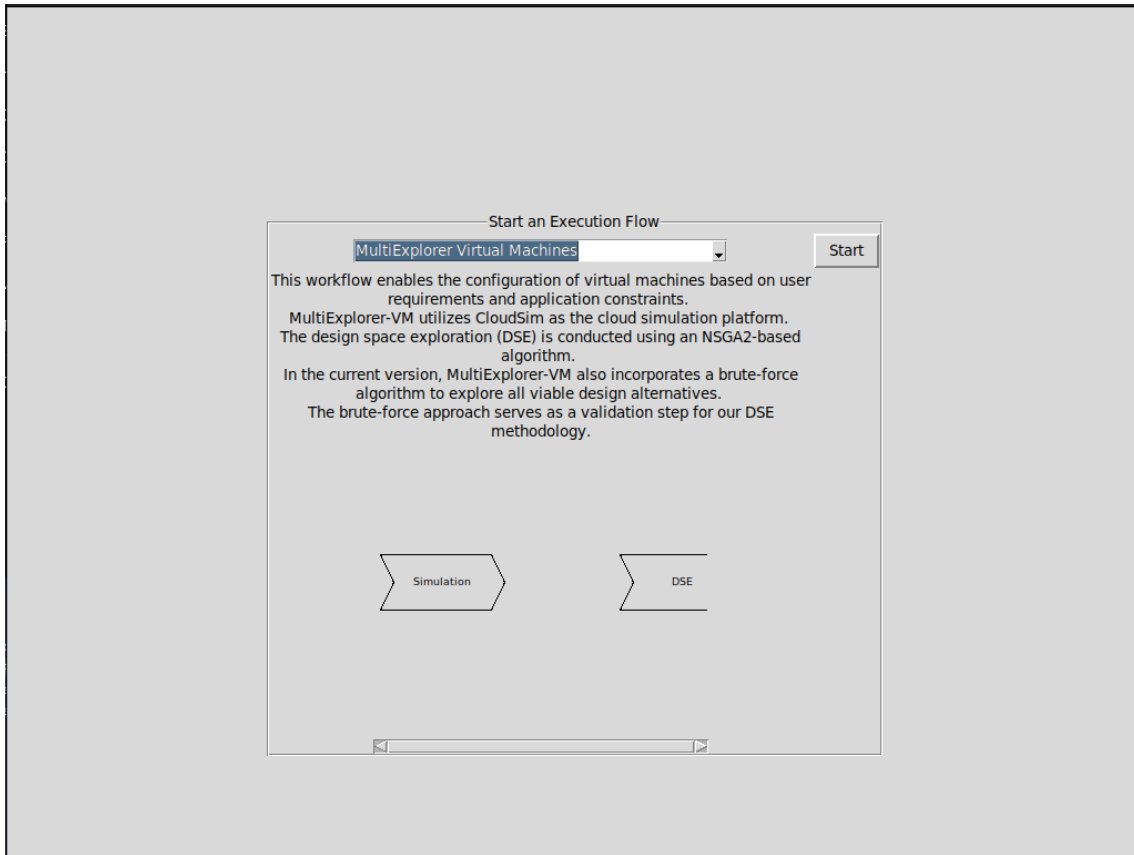


Figure 11: Virtual machines's execution flow.

For the virtual machines flow, it first shows a description of each step that it goes through (Figure 11).

MultiExplorer Virtual Machines Settings

Simulation DSE

Application

Applicaton VM EP-S

Model VM m5n.large

Cores Cloud set

Cores Cloudlet for design 32

Constraints

Maximum Cost 2.0 (USD/h)

Maximum Time 1.0 (hours)

Back Execute

Figure 12: User inputs for simulation step.

Now, the user must select the benchmark application and the initial configuration model - an AWS (Amazon Web Services) instance. Additionally, the user must specify the number of cores required by the cloudlet, a class that models cloud-based applications and services commonly used in data centers. To select the recommended configurations, the DSE algorithm will require the necessary constraints, as shown in Figure 12.

MultiExplorer Virtual Machines Settings

Simulation

DSE

Exploration Space

Run brute force as well ☒

Suplementar number VM for design - from 1 to 3

Original number VM for design - from 1 to 3

NSGA-II Parameters

Crossing Rate 50.0 (%)

Mutation Rate 10.0 (%)

Number of Generations 50

Population Size 20

Back

Execute

Figure 13: User inputs for DSE step.

For the DSE settings (Figure 13), the user must specify the quantity of each original configuration and additional model, which together represent a heterogeneous combination. The primary exploration algorithm is NSGA-II - a genetic algorithm - but the user also has the option to run a brute force algorithm by checking the 'Run brute force as well' box. Finally, the default parameters for the NSGA-II algorithm can be modified.



Figure 14: processing window.

Figure 14 shows the processing window. In this window, you can view a preview of the results once the process is complete.

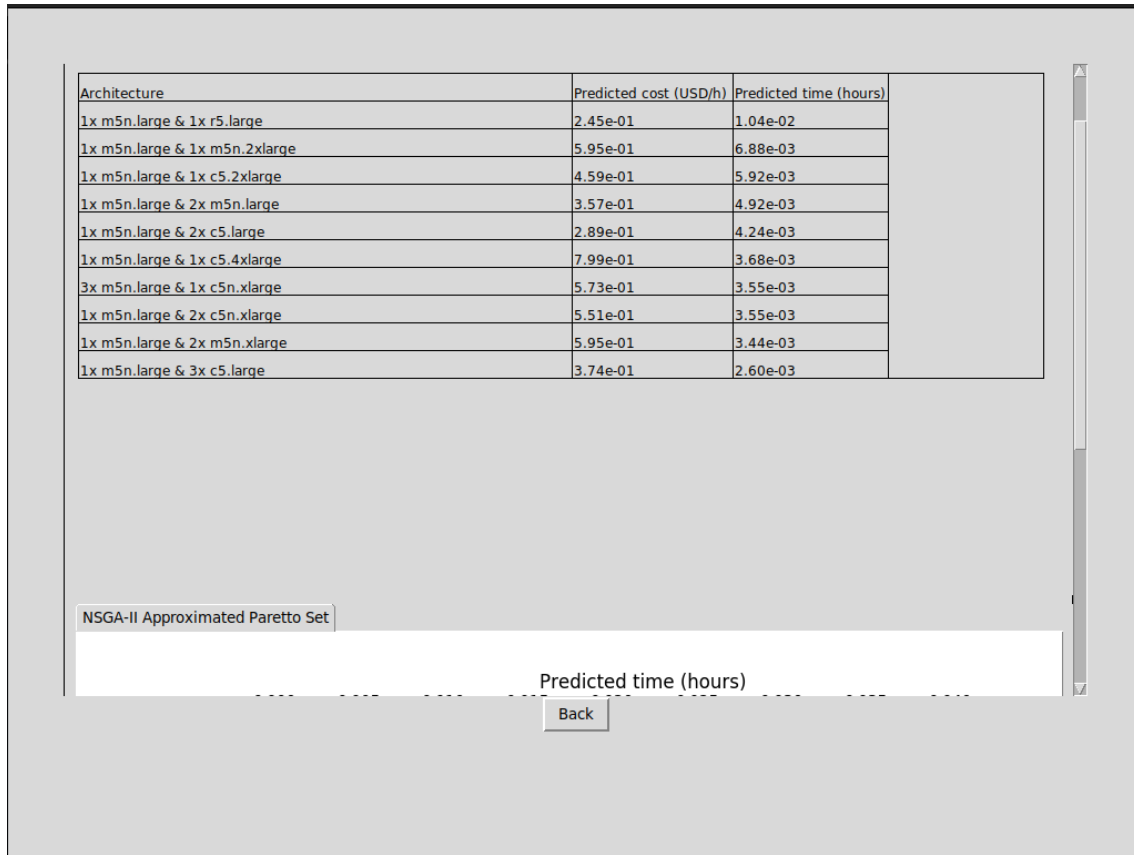


Figure 15: Recommended configurations displayed in tables and graphs.

The final window displays the recommended configurations (Figure 15) in the form of a table and graphs. If the brute force algorithm was selected, its configurations will also be displayed.

2.3 MultiExplorer GPUs

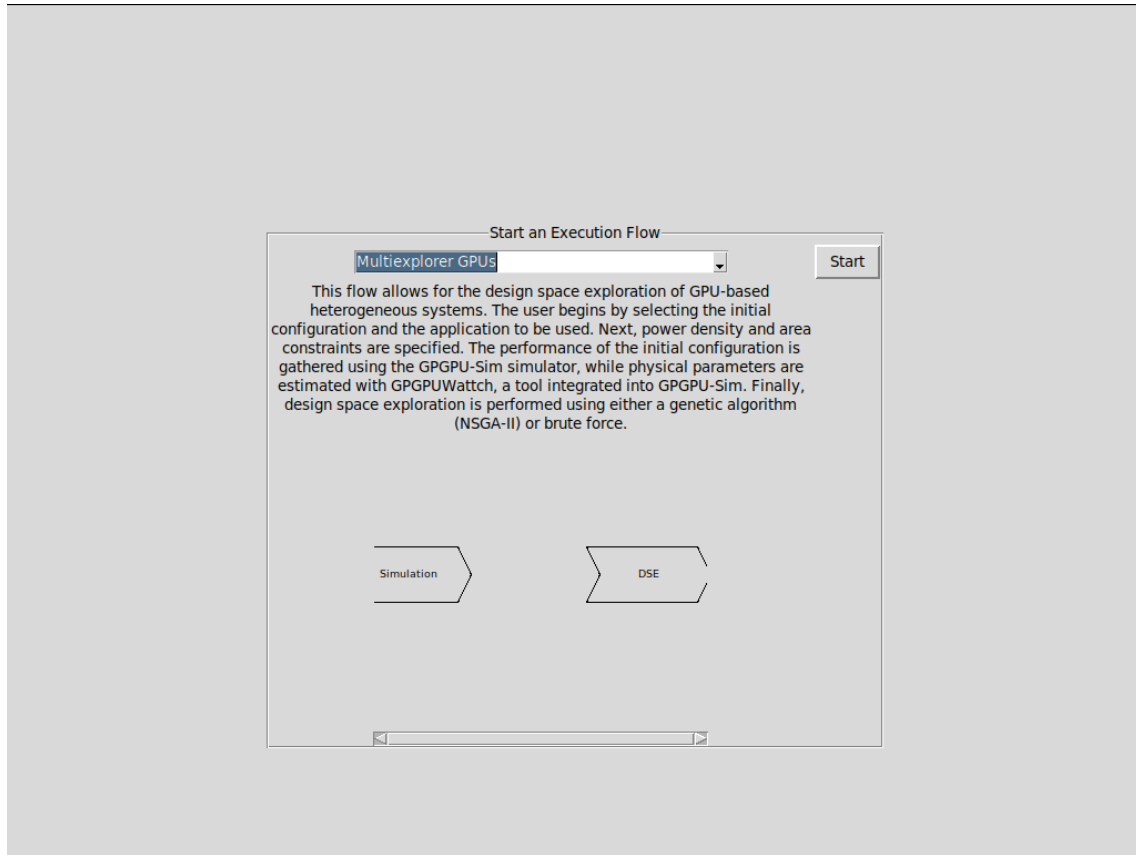


Figure 16: MultiEXplorer GPUs execution flow.

After selecting the "Multiexplorer GPUs" flow, the description of the steps will be displayed as shown in Figure 16."

The image shows a software window titled "Multiexplorer GPUs Settings". At the top, there are two tabs: "Simulation" and "DSE", with "DSE" being the active tab. Below the tabs is a large rectangular area containing a "Model Settings" section. This section has two dropdown menus: "Application" with the value "asynAPI" and "Model" with the value "GTX_480". At the bottom of the window, there are two buttons: "Back" on the left and "Execute" on the right.

Figure 17: User inputs for simulation step.

For the simulation step (Figure 17), the user must select the application to be used in the initial configuration (Model) simulation. Additional parameters for the GPU models can be found in `"/multiexplorer/input-examples/GPUs"`.

Multiexplorer GPUs Settings

Simulation
DSE

Exploration Space

GPU Cores for design - from
1
to
20

Original Cores for design - from
1
to
15

NSGA-II Parameters

Crossing Rate
50.0
(%)

Mutation Rate
10.0
(%)

Number of Generations
100

Population Size
10

DSE Settings

Run brute force aswell
☒

Constraints

Maximum Power Density
0.3
(V/mm²)

Maximum Area
200
(mm²)

Back
Execute

Figure 18: User inputs for DSE step.

For the DSE settings (Figure 18), the interval for the quantity of *Streaming Multiprocessors* (SMs) must be specified for both the initial configuration and the heterogeneous combination model. After that, the default NSGA-II parameters can be modified. The user can also run a brute force algorithm in addition to the genetic algorithm (NSGA-II), which tries all possible combinations within the selected intervals. For the recommended configurations, constraints such as chip area and maximum power density must be specified.



Figure 19: Processing window.

Figure 19 shows the processing for each step. After a step is completed, the user can view preview results for that step, such as performance metrics from the simulator, for example.

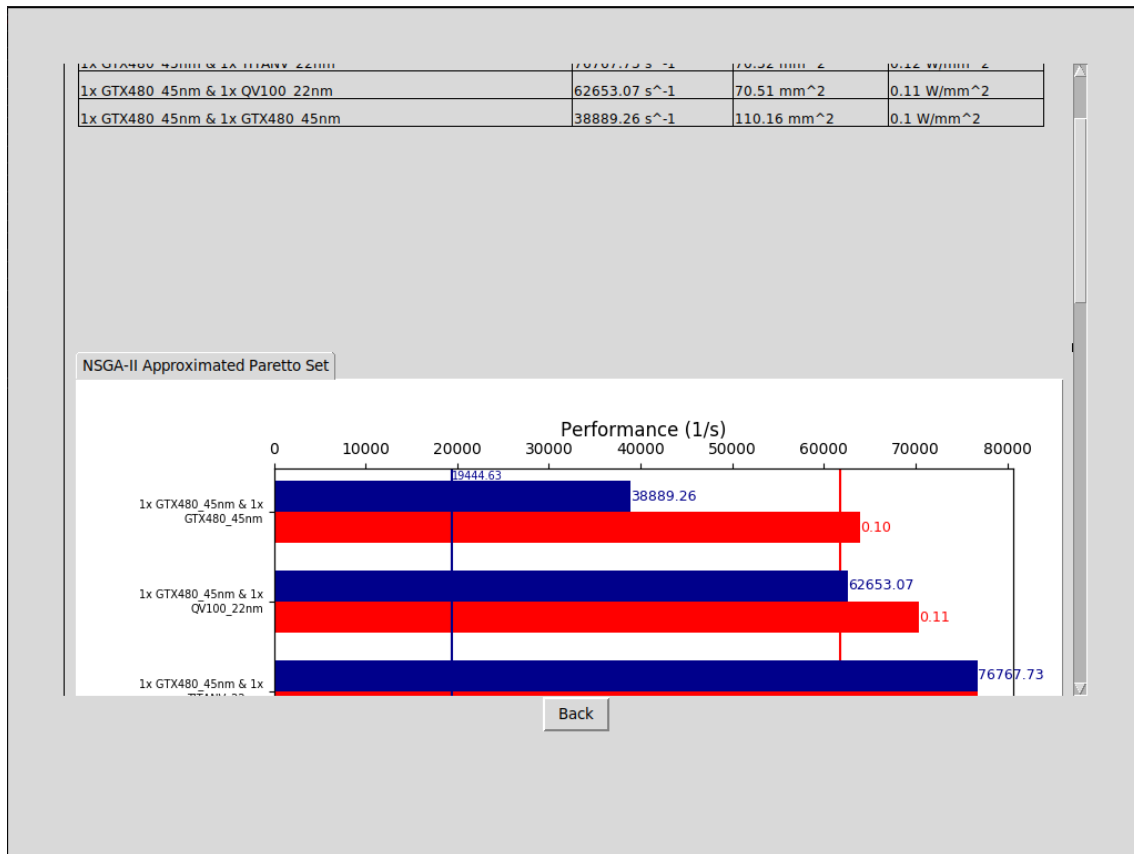


Figure 20: Results window.

The final window will display the recommended configurations, which combine the initial configuration and models from the database. These configurations will be shown in tables and graphs (Figure 20).